THERMOR-R: The Heatsink That Grows When It Gets Hot

A passive, self-deploying cooling system

The Problem:

A heatsink is always a compromise between performance and size. A large heatsink can handle high thermal loads but becomes bulky and costly, while a compact heatsink may struggle during sudden increases in power dissipation. This can affect chargers, LED lighting and power electronics. THERMOR-R explores whether a heatsink can remain compact during normal operation and automatically expand its cooling surface when thermal conditions demand it.

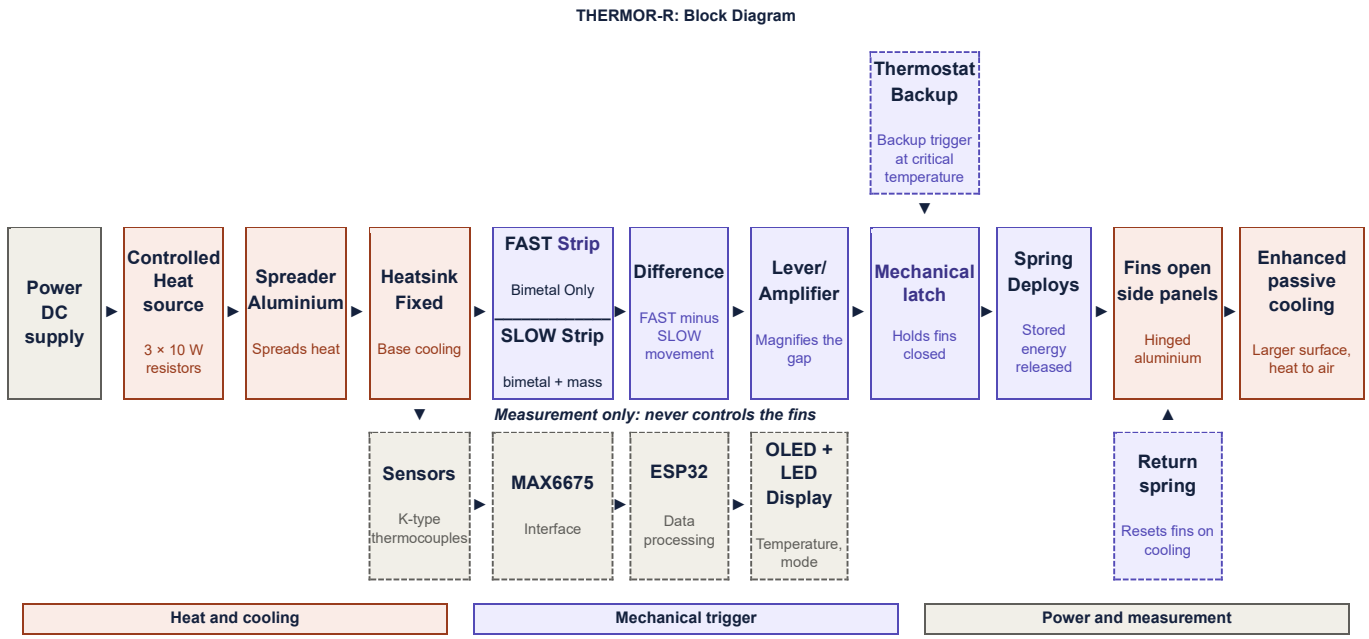
Our Idea:

THERMOR-R combines a fixed central heatsink with two folded side fin panels that automatically open during a rapid temperature rise without motors, software or human intervention. The key mechanism uses two thermal elements with different response rates, allowing heat itself to trigger the mechanical cooling response.

Methodology:

1. Heat source: Three 10 W power resistors mounted on an aluminium plate provide controlled heat to the central heatsink. A FAST bimetal strip responds rapidly to temperature rise, while a SLOW strip with added thermal mass responds more gradually. During rapid heating, the difference in their movement creates a gap.
2. Mechanical trigger: This movement is transferred to a mechanical lever, which amplifies it and releases a latch. A loaded spring then swings the hinged side fins open, increasing the exposed cooling surface.
3. Backup: A separate critical-temperature thermostat provides a backup trigger if the temperature becomes high even when the FAST-SLOW difference is insufficient.
4. Reset: A return spring folds the fins back as the system cools.
5. Monitoring: K-type thermocouples, MAX6675 modules, an ESP32 and OLED monitor temperature and operating mode; they only observe and do not control the fins.
6. Proof: THERMOR-R will be compared with a conventional fixed heatsink under identical heat input. Four tests—normal heating, sudden thermal surge, slow but high heating, and cooling—will be performed. Temperature-versus-time, peak temperature, response time and recovery behaviour will be recorded to evaluate whether the adaptive configuration provides measurable thermal improvement**.**

Proposed Block Diagram:



Dashed boxes: backup trigger, return spring and measurement-only blocks.

Proposed Outcome

1. Fins stay closed in normal use and open automatically on a thermal surge or high temperature.
2. Lower peak temperature and faster recovery than a fixed heatsink of the same footprint, shown with real graphs.
3. A transparent ~250 × 180 × 150 mm tabletop model where judges can watch closed → trigger → open.
4. A low-cost starting point for compact, self-adjusting cooling in LED lighting, chargers and EV electronics.

"A passive heatsink that changes its own cooling capacity when thermal conditions demand it."